Enrollment No._____

GANPAT UNIVERSITY

B.Sc. IT (Cyber Security) Semester -III(CBCS)

APRIL-JUNE 2024 Remedial

U63A2DMS DATABASE MANAGEMENT SYSTEM

Time: 3 Hours Total: 60 Marks

Instructions:

1. Figures to the right indicate full marks, Course Outcomes and BL Pattern Level.
2. This Question paper has two sections. Each section should be written in a separate answer book.
3. Section-I consist of 07 questions of 6 marks each with options respectively. Section-II consist of 03 questions, Question 2 of 12 Marks, Question 3 of 12 Marks and Question 4 of 6 Marks with options respectively.
4. Be precise and to the point in your answer.
5. CO is Course Outcome from the Subject Syllabus.
6. Bloom's Taxonomy Based Assessment Pattern: BTL1-Remember; BTL2-Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create.

SECTION – I

Q-1	Answer the following. (Any 5)	Marks 30	CO	Bloom's Taxonomy Pattern
1	Elaborate the three-tier architecture.	06	CO1	BTL4
2	Analyze the responsibility of database administration.	06	CO1	BTL4
3	Evaluate pattern matching operator in detail.	06	CO3	BTL5
4	classify the DML SQL command in details.	06	CO3	BTL4
5	Apply Trivial and Non-Trivial dependency with example.	06	CO2	BTL3
6	Evaluate Subquery in SQL in details.	06	CO4	BTL5
7	Demonstrate index and view in details.	06	CO5	BTL3

SECTION – II

Q-2	Answer the following (Any Two)	Marks 12	CO	Bloom's Taxonomy Pattern
1	Illustrate on Dr. E. F. Codd rules.	06	CO1	BTL3
2	What is E-R diagram? Explain components of it.	06	CO2	BTL5
3	What is normalization? Explain Boyce/Codd normalization in details.	06	CO2	BTL2
Q.3	**Answer the following: (Any Two)**	**12**		
1	Define constraints? Explain primary key and foreign key in detail.	06	CO3	BTL2
2	Describe different types of Joins in SQL.	06	CO4	BTL2
3	Calculate string Functions with example.	06	CO4	BTL3
Q.4	**Answer the following: (Any One)**	**06**		
1	Evaluate the spooling concept in details.	06	CO5	BTL5
2	Design normalizes database design with ER relationship of college management system.	06	CO5	BTL6

**** End of the Paper ****

Enrollment No._____

GANPAT UNIVERSITY

B.Sc.IT(CS) - III (CBCS)

APRIL-JUNE 2024 Remedial

U63A3OCP: OBJECT ORIENTED CONCEPTS AND PROGRAMMING

Time: 3 Hours | Total: 60 Marks

Instructions:

1. Figures to the right indicate full marks, Course Outcomes and BL Pattern Level.
2. This Question paper has two sections. Each section should be written in a separate answer book.
3. Section-I consist of 07 questions of 6 marks each with options respectively. Section-II consist of 03 questions, Question 2 of 12 Marks, Question 3 of 12 Marks and Question 4 of 6 Marks with options respectively.
4. Be precise and to the point in your answer.
5. CO is Course Outcome from the Subject Syllabus.
6. Bloom's Taxonomy Based Assessment Pattern: BTL1-Remember; BTL2-Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create.

Section – I

Q-1	Answer the following. (Any 5)	Marks 30	CO	Bloom's Taxonomy Pattern
1	Explain the fundamental principles of OOP.	06	CO2	BTL2
2	Describe the differences between the String and the StringBuffer in Java.	06	CO1	BTL4
3	Explain the different types of inheritance.	06	CO2	BTL5
4	Provide primary differences between List and Set in the Java Collection Framework.	06	CO3	BTL4
5	What is InetAddress in Java? Can you provide a simple example of how to obtain and use InetAddress in a Java program?	06	CO3	BTL6
6	Explain URLConnection and HttpURLConnection.	06	CO3	BTL1
7	Describe the process of passing Lambda Expressions as arguments to methods in Java with example.	06	CO1	BTL1

Section – II

Q-2	Answer the following (Any Two)	Marks 12	CO	Bloom's Taxonomy Pattern
1	Describe the use of the "final" keyword in Java and its implications for classes, methods, and variables	06	CO2	BTL3
2	Explain how interfaces are defined and implemented and discuss the benefits of using interfaces.	06	CO2	BTL5

3	Discuss the advantages and challenges associated with multithreading and provide an example demonstrating the creation and execution of multiple threads.	06	CO1	BTL6
Q.3	**Answer the following: (Any Two)**	**12**		
1	How do you implement and work with generic types in Java, and how does it contribute to code reusability and type safety?	06	CO2	BTL5
2	Explain the Hashtable class in Java and how it differs from other data structures in the java.util package.	06	CO3	BTL1
3	What is the role of Server Sockets in Java for TCP/IP communication? Explain with example.	06	CO3	BTL3
Q.4	**Answer the following: (Any One)**	**06**		
1	Explain the concept of Lambda Expressions in Java, and how they are used to simplify code. Provide an example of a Block Lambda Expression and discuss when it is preferable over a simple Lambda Expression.	06	CO1	BTL4
2	How do Lambda Expressions in Java handle exceptions? Discuss the syntax and techniques for handling exceptions within Lambda Expressions.	06	CO1	BTL5

**** End of the Paper ****

Enrollment No._____

GANPAT UNIVERSITY

B.Sc.IT(CS) – III (CBCS)

Remedial – APRIL-JUNE 2024

U63A4CN1: COMPUTER NETWORK-I

Time: 3 Hours

Total: 60 Marks

Instructions:

1. Figures to the right indicate full marks, Course Outcomes and BL Pattern Level.
2. This Question paper has two sections. Each section should be written in a separate answer book.
3. Section-I consist of 07 questions of 6 marks each with options respectively. Section-II consist of 03 questions, Question 2 of 12 Marks, Question 3 of 12 Marks and Question 4 of 6 Marks with options respectively.
4. Be precise and to the point in your answer.
5. CO is Course Outcome from the Subject Syllabus.
6. Bloom's Taxonomy Based Assessment Pattern: BTL1-Remember; BTL2-Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create.

Section – I

Q-1	**Answer the following. (Any Five)**	**Marks 30**	**CO**	**Bloom's Taxonomy Pattern**
1	What is computer network? What are the advantages and usages of computer networking? Discuss.	06	CO1	BTL1, BTL2
2	What is subnetting? Justify the statement " subnetting can improve your networking experience."	06	CO3	BTL5
3	What is routing? Explain three types of routing and distinguish static and dynamic routing.	06	CO3	BTL4
4	What are the different ways to secure a computer network? How do you implement and evaluate various strategies to enhance security of a computer network?	06	CO5	BTL3, BTL6
5	What is address resolution protocol (ARP)? How does it work? Discuss.	06	CO4	BTL2, BTL3
6	What is firewall? Demonstrate the work of firewall.	06	CO5	BTL4
7	Demonstrate OSI reference model.	06	CO1	BTL4

Section – II				
Q-2	**Answer the following: (Any Two)**	**Marks 12**	**CO**	**Bloom's Taxonomy Pattern**
1	Explain different types of computer network.	06	CO1	BTL2
2	What is network topology? Explain and compare different network topologies.	06	CO1	BTL2 ,BTL4
3	What is switching? Explain different types of switching techniques.	06	CO2	BTL2, BTL4
Q-3	**Answer the following: (Any Two)**	**12**		
1	Differentiate the MAC address with the IP address.	06	CO3	BTL4
2	Differentiate TCP with UDP protocol.	06	CO4	BTL4
3	Define: 1) Unicasting 2) Anycasting 2) Multicasting 4)Broadcasting	06	CO3	BTL1
Q-4	**Answer the following: (Any One)**	**06**		
1	What is Encryption? Why do we need Encryption? How does Asymmetric Encryption works?	06	CO5	BTL2, BTL4
2	What is computer virus? Assess the ethical implications of creating and distributing computer viruses. What are the potential consequences, both legally and socially, for individuals involved in such activities?	06	CO5	BTL5

**** End of the Paper ****

Enrollment No._____

GANPAT UNIVERSITY

B.Sc.IT(CS)– SEM-III EXAMINATION (CBCS)

APRIL-JUNE 2024 Remedial

U63A5DF1: DIGITAL FORENSICS -I

Time: 3 Hours

Total: 60 Marks

Instructions:

1. Figures to the right indicate full marks, Course Outcomes and BL Pattern Level.
2. This Question paper has two sections. Each section should be written in a separate answer book.
3. Section-I consist of 07 questions of 6 marks each with options respectively. Section-II consist of 03 questions, Question 2 of 12 Marks, Question 3 of 12 Marks and Question 4 of 6 Marks with options respectively.
4. Be precise and to the point in your answer.
5. CO is Course Outcome from the Subject Syllabus.
6. Bloom's Taxonomy Based Assessment Pattern: BTL1-Remember; BTL2-Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create.

Section – I				
Q-1	**Answer the following. (Any 5)**	**Marks 30**	**CO**	**Bloom's Taxonomy Pattern**
1	Describe the role of digital forensics in the incident response process	06	CO1	BTL1
2	Differentiate between an incident response framework and an incident response plan.	06	CO1	BTL3
3	Identify and discuss the essential tools and hardware used in digital forensics.	06	CO2	BTL2
4	Elaborate importance of network diagrams	06	CO3	BTL6
5	Describe the steps involved in acquiring host-based evidence	06	CO4	BTL5
6	Compare dead imaging and live imaging	06	CO5	BTL4
7	Discuss the process of forensic imaging using Autopsy	06	CO3	BTL5
Section – II				
Q-2	**Answer the following (Any Two)**	**Marks 12**	**CO**	**Bloom's Taxonomy Pattern**
1	Define the incident response process	06	CO1	BTL6
2	Describe the components of a digital forensics' lab and the importance of physical security	06	CO2	BTL4
3	Explain the laws and regulations of digital forensics	06	CO2	BTL2
Q.3	**Answer the following: (Any Two)**	**12**		

1	Discuss the process of collecting packet capture as network evidence	06	CO3	BTL2
2	Evaluate the importance of memory acquisition in host-based evidence collection.	06	CO4	BTL3
3	Explain the significance of security information and event management (SIEM) system.	06	CO3	BTL4
Q.4	**Answer the following: (Any One)**	**06**		
1	Diff b/w Volatile and Non Volatile Memory.	06	CO5	BTL3
2	Illustrate Forensic Report	06	CO5	BTL5

**** End of the Paper ****